Reg. No.:	
Name :	

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MID TERM EXAMINATIONS – September 2023

Programme	:	**B.Tech.**	Semester	:	**Fall 2023-24**
Course Title /Course code	:	**Introduction to computational physics / PHY1003**	Slot	:	**D11+D12+D13+D14**
Time	:	**1 ½ hours**	Max. Marks	:	**50**

Answer all the Questions

Q.No.	Sub. Sec.	Question Description	Marks
1	(a)	With the help of appropriate example (include the flow chart for the example) explain flow control structures in the programming language.	**7**
1	(b)	A company conducts a survey to rate a product based upon the price and quality. Classify the data thus obtained and methods to analyse it.	**3**
2		x: 0, 0.5, 1.0, 1.5, 2.0, 2.5 y: -0.4326, -0.1656, 3.1253, 4.7877, 4.8535, 8.6909 Evaluate the function f(x) for a linear fit of the above given data.	**10**
3		Consider a set of linear equations: $x_1 + 2x_2 + x_3 = 0$ $2x_1 + 2x_2 + 3x_3 = 0$ $-x_1 - 3x_2 = 2$ Solve the system of the linear equation utilizing the Gauss-Jordon elimination.	**10**
4		For the system of linear equations given in question 3, draw a flowchart outlining the structure of a computer program to obtain its solutions.	**10**
5		Illustrate the origin of the error in computational calculation of function 'e^x'. For x=1, tabulate the estimated value, absolute error and percentage error for n=3, 4, and 5. Assume the e = 2.7182812.	**10**

Data table for Question 2:

x	0	0.5	1.0	1.5	2.0	2.5
y	-0.4326	-0.1656	3.1253	4.7877	4.8535	8.6909

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